Econ 131
Spring 2026
Danny Yagan

Problem Set 3

DUE: May 1, 2026, at 5PM PT on Gradescope

The problem set is due on Gradescope at 5pm sharp on the due date, and solutions will be posted at noon the next day or shortly thereafter. If you submit it to Gradescope between the 5pm deadline and the noon posting, you can get up to half credit. (If you have a DSP accommodation for extra time on take-home assignments, you will get up to full credit if you submit it by the noon deadline.) No submissions after the noon deadline will be accepted for any student, regardless of when solutions are posted.

Student Name: Marcus Huang

Student ID: 3041432436

- Write or type your answers clearly and in dark ink (physical or electronic ink) so that your responses are legible.
- Tag each of your answers on Gradescope so that it is clear what responses are to which questions.
- **Although you may work in groups**, each student must submit individual sets of solutions. You must note the names of other students that you worked with. Write their names here:

1. True/False/Uncertain with Explanation [30 points]

Determine whether each statement is true, false, or uncertain and explain why. Answers with no explanation will receive no points. [5 points each]

(a) If schooling is a screening device, the government should encourage more education.

False. If schooling is a screening device, education does not increase productivity, it merely allows high-ability workers to signal their pre-existing ability to employers. While the private return to education is positive (higher wages), the social return is zero, because no real output is created. Subsidizing more education would simply raise the equilibrium signaling threshold without generating any net social value, wasting real resources in a credential arms race. Therefore, the government should not encourage more education under the pure screening model.

(b) In the standard expected-utility model of unemployment insurance, if higher UI benefits do not change job-search behavior or unemployment durations, full insurance between employment and unemployment states is efficient.

True. Optimal UI balances consumption smoothing (benefit) against moral hazard (cost). If UI has no behavioral effects, the moral hazard cost is zero, so the only consideration smoothing. A risk-aversion individual maximizes expected untility when consumption is equalized across employed and unemployed states, exactly what full insurance achieves, making it efficient.

(c) Suppose that Obamacare had done only one thing: made it illegal for health insurance providers to deny applicants based on preexisting conditions ("community rating"). This policy would have reduced adverse selection.

False. Community rating alone would worsen, not reduce, adverse selection. By forcing a uniform premium, it makes insurance attractive to high-risk individuals and unattractive to low-risk individuals. Healthy people drop out, the pool gets sicker, premiums rise further, a classic adverse selection death spiral. Reducing adverse selection requires pairing community rating with something that keeps healthy people in the pool, like an individual mandate.

(d) Evidence that more generous unemployment insurance increases unemployment durations implies that the optimal unemployment insurance benefit must be zero.

False. Evidence of more hazard only establishes the cost side of the UI tradeoff, it says nothing about the benefit side. The optimal benefit equates the marginal consumption smoothing benefit to the marginal moral hazard cost. Near zero benefits, the insurance value is large while the destortion is small, so the optimum is strictly positive. Moral hazard evidence shifts the optimal benefit downward from full insurance but does not drive it to zero.

(e) If you believe that the moral hazard cost of disability insurance identified by judges-design evidence is larger than its social benefit for marginal applicants, then you should want to eliminate the disability insurance program altogether.

False. The judges-design evidence identifies costs and benefits only for maginal applicants, borderline cases near the eligibility threshold. Even if costs exceed benefits at the margin, clearly disabled inframarginal applicants still receive large net benefits from DI. The correct policy response is to tighten eligibility to screen out marginal cases, not eliminate a program that still substantially benefits the severely disabled.

(f) Evidence that reported workers' compensation claims for hard-to-verify back injuries spike on Mondays is consistent with moral hazard.

True. Back injuries are hard to verify, so workers can strategically report weekend injuries as Monday workplace injuries to collect workers' compensation. The Monday spike, absent in easily verified injuries, suggests workers are responding to insurance coverage by misreporting injury origins. This behavioral distortion cased by the existence of insurance is precisely what economists define as moral hazard.

2. Empirical Problem: High Point University and the Top 20% [30 points]

We have uploaded a PDF of the Wall Street Journal article "How a Small North Carolina College Became a Magnet for Wealthy Students" with the problem set materials; the article is also available here. For the data, use the Opportunity Insights repository here and the Chetty-Friedman-Saez-Turner-Yagan (2020) college mobility paper here.

We want to determine whether the January 2005 arrival of High Point president Nido Qubein changed the share of High Point students with top-20%-income parents. In the longitudinal file, `cohort` is birth cohort, so age-20 year is `cohort+20`, and `par_q5` is the share of students with top-20%-income parents. Because students first admitted under Qubein in fall 2005 would typically appear in the age-20 data in 2007, treat 2007 as the first affected age-20 cohort, draw the vertical line in your event-study graph at 2006.5, let $HIGHPOINT_c = 1$ if college c is High Point University and $HIGHPOINT_c = 0$ otherwise, and let $POST_t = 1$ for $t \geq 2007$ and $POST_t = 0$ otherwise.

As a baseline control group, use colleges in Virginia, South Carolina, Georgia, and Tennessee that are private, non-profit, four-year institutions and classified as "Selective private" (tier 6) in the Opportunity Insights college-characteristics file, the same tier as High Point. This is a reasonable but imperfect starting point because it compares High Point to nearby schools in the same broad market segment, so you should assess it with the graph.

Use one observation per college-year and standard (i.e., unweighted ordinary least squares) regressions throughout. Do not worry about small-sample statistical inference issues like large standard errors; focus on the design, the graph, and the coefficient estimates. You may complete the empirical work in a notebook using `https://datahub.berkeley.edu/`, or use AI tools. In your submitted solution file, include screenshots of the relevant notebook output (graph and regression results) alongside your written answers.

(a) [4 points] From the Opportunity Insights data repository, download the following three files under "Mobility Report Cards":

i. "Preferred Estimates of Access and Mobility Rates by College" (Table 1)

ii. "Baseline Longitudinal Estimates of Child and Parent Income Distributions by College and Child's Cohort" (Table 3)

iii. "College Level Characteristics from the IPEDS Database and the College Scorecard" (Table 10)

Identify High Point University in these files. Report High Point's `super_opeid`, state, sector, and tier.

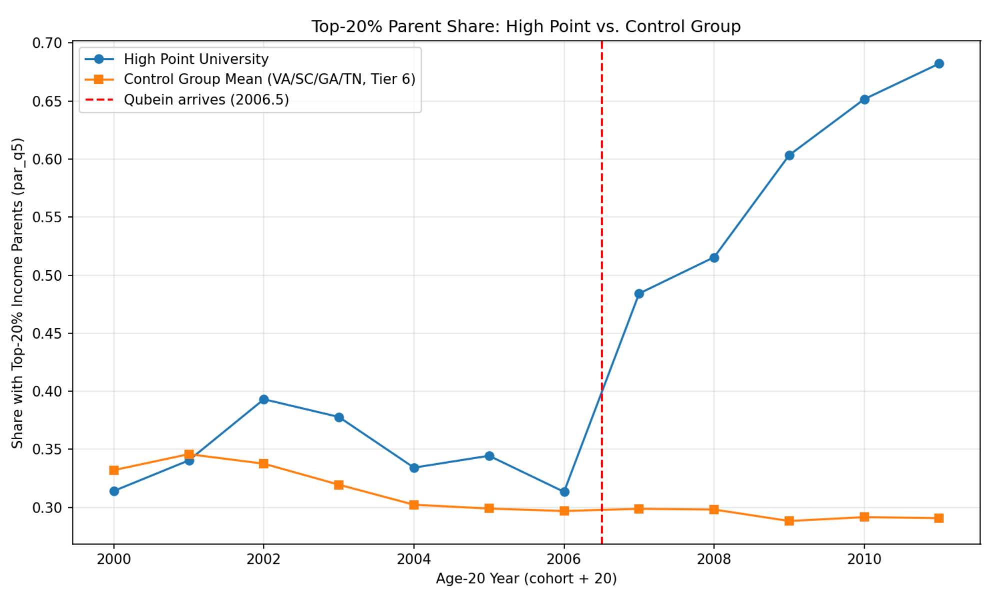
- High Point's super_opeid : 2933
- state : North Carolina
- sector : Private non-profit
- tier : 6 (Selective private)

(b) [5 points] Using the college-characteristics file, construct the baseline control group described above. How many colleges are in your control group? Are the control-group colleges relatively evenly divided across the four states, or are they mostly concentrated in a single state?

Using the corrected variable deffinitions for colleges in VA. SC. GA, and TN, the control group contains 67 colleges. The colleges are evenly distributed in state Y. suggesting the control group draws broadly from one part of the region.

(c) [7 points] Using the longitudinal file, create a graph with the share of students with top-20%-income parents on the y-axis and year on the x-axis. Use `par_q5` for the y-variable and `cohort+20` for the x-variable, where the x-axis year denotes the year that the students turn age 20. Plot two lines: one line for High Point, and a second line that is the simple unweighted mean value for the control-group colleges. Draw a vertical line at 2006.5. Do the treatment and control groups appear to exhibit parallel trends before 2007?

The graph plots par_q5 for High Point and the control group mean from 2000 to 2011, with a vertical line at 2006.5. Before 2007, both lines fluctuate in a similar range (roughly 0.3-0.39), moving broadly in parallel, the parallel trends assumption appears approximately satisfied. After 2007, High Point's top 20% parent share rises sharply from 0.31 to 0.68 by 2011, while the control group remains flat near 0.29-0.3. This dramatic post-2007 divergence is consistent with Qubein's arrival causing a major shift in the socioeconomic coposition of High Point's student body.



(d) [6 points] Estimate the overall difference-in-differences specification

$$y_{ct} = \alpha_c + \gamma_t + \delta\,(HIGHPOINT_c \times POST_t) + \varepsilon_{ct},$$

where y_{ct} is the top-20% parent share for college c in age-20 year t, α_c are college fixed effects, and γ_t are year fixed effects. Report and interpret $\hat{\delta}$.

$\hat{\delta} = 0.2625$, this means that after 2007, High Point's share of students with top-20% income parents increased by 26.25% more than the control group, after removing the pre-existing level difference between High Point and control school (β) and the common post-2007 time trend across all schools (γ).

(e) [4 points] Estimate a "2011 DD" effect by restricting the sample to the pre period ($t <$ 2007) and 2011, and then redefining $POST_t = 1$ for $t = 2011$ and $POST_t = 0$ for $t < 2007$. Run

$$y_{ct} = \alpha_c + \beta POST_t + \delta \left(HIGHPOINT_c \times POST_t \right) + \varepsilon_{ct}.$$

Report and interpret $\hat{\delta}$.

$\hat{\delta} = 0.3602$, this means that by 2011 specifically, High Point's top-20% parent share was 36.02% higher than in the pre-period relative to the control group.

(f) [4 points] Based on these empirical results and on what we discussed in class about how financial aid works, what might President Qubein's long-run financial strategy have been? There is no right answer here, but give a reasonable speculation. If it helps to have an example, think about the Wall Street Journal article that touted very expensive amenities like the planetarium, which was built in 2018 and cost millions of dollars.

Qubein's likely long-run financial strategy was to reposition High Point as a luxury "amenty college" that attracts full-paying wealtly students. Since wealthy families do not qualify for need-based financial aid, recruiting more top-20% income students directly raises net tuition revenue per student. Qubein then reinvested that revenue into high-visibility luxury amenties, like the multimillion-dollar planetarium, which furth attract affluent applicants willing to pay full sticker price. This creates a self-reinforcing cycle: more wealtly effectivly uses price discrimination in reverse, rather than giving aid to attract lower-income students, it uses amenties to attract higher-income students who subsidize the institution's growth.

3. Analytical Question: Stylized Unemployment Insurance [20 points]

A government is designing an unemployment insurance program. The individual earns wage $w = \$10$ if employed and earns \$0 if unemployed. The government offers benefit b if unemployed, financed by a tax p levied regardless of employment status. Let s denote the probability that the individual is unemployed and claims benefits; she can choose $s \in [0.01, 0.99]$.

Crucially, the individual can influence s through her own effort, such as searching harder for work. This is the source of *moral hazard*: the generosity of unemployment insurance may itself affect behavior. The individual's expected utility is:

$$E[U] = (1-s)\ln(w-p) + s\ln(b-p) - (1-s)^2$$

Here $w - p$ is consumption if employed, $b - p$ is consumption if unemployed, and the final term $-(1-s)^2$ captures the increasing personal cost of reducing s.

Note: Parts (g) through (k) can be completed successfully without using the answers from parts (a) through (f). For questions that depend on answers to previous parts, you can still get full credit if your logic is correct even if your answer to a previous part is wrong. Therefore, be sure to explicitly state your logic.

(a) [2 points] The government first considers Program #1, which provides *full insurance*—equalizing consumption across employment and unemployment. Suppose the government believes the probability of unemployment is 10%. What benefit b and tax p make this program budget balanced? *Hint:* expected tax revenue is p, and expected benefit payments are sb.

$$w - p = b - p \Rightarrow b = w = \$10$$

$$\Rightarrow p = s \cdot b = 0.1 \times 10 = \$1$$

under P#1, full insurance requires $b = \$10$. With the government's assumed unemployment probability of 10%, budget balance requires $p = \$1$. Both the employed and unemployed consume \$9.

(b) [2 points] Write out the individual's optimization problem under Program #1. Solve for her optimal choice of s. How does this compare to the government's assumed unemployment probability of 10%?

Under P#1, since $b - p = w - p = 9$, consumption is identical in both states, so the utility function simplifies to:

$$E[U] = \ln(9) - (1-s)^2$$

which is strictly increasing in s. the individual maximizes utility by choosing the highest feasible $s^* = 0.99$, far above the government's assumed 10%. With full insurance, there is no consumption benefit from working, so the individual exerts no job-search effort.

(c) [1 point] What is the individual's expected utility under Program #1?

$$E[U] = \ln(9) - (1-0.99)^2 = \ln(9) - (0.01)^2 = \ln(9) - 0.0001$$

$$\Rightarrow \ln(9) \approx 2.1972$$

$$\Rightarrow E[U] \approx 2.1972 - 0.0001 \approx 2.197$$

(d) [2 points] Suppose the government keeps the tax at $p = 1$ from Program #1 even after the individual's behavioral response from part (b). Does Program #1 run a surplus, break even, or run a deficit? Explain why this is an example of moral hazard rather than adverse selection.

With $p = \$1$ and actual $s = 0.99$, tax revenue is \$1 but expected benefit payments are $0.99 \times \$10 = \$9.9$, so P#1 runs a deficit of \$8.9. This is moral hazard, not adverse selection, because the individual is not a different type from what the government assumed; rather, she changed her job-search behavior in response to the full-insurance terms. Once consumption is equalized across states, there is no incentive to search for work, so she maximizes unemployment.

(e) [1 point] Based on the social-insurance logic from lecture, does the fiscal imbalance in part (d) suggest that full insurance is likely to be optimal in this setting? Briefly explain the intuition.

No, the fiscal imbalance strongly suggests full insurance is not optimal here. The deficit of \$8.9 reveals a massive moral hazard cost: the individual raised her unemployment probability from the assumed 10% to 99% once consumption was fully equalized. The social insurance logic says optimal UI balances consumption smoothing benefits against moral hazard cost. When moral hazard is this severe, the optimal benefit must be set below full insurance to preserve job-search incentives.

(f) [2 points] Suppose the government nevertheless insists on providing full insurance, but adjusts the tax to keep the program budget balanced after accounting for the individual's behavioral response from part (b). What tax is required? What is the individual's expected utility under this budget-balanced full-insurance program? Call this **Program #2**.

To maintain full insurance ($b = \$10$) while balancing the budget with actual $s = 0.99$, the required tax is $p = 0.99 \times \$10 = \$9.9$

This leaves consumption of only $10 - 9.9 = \$0.1$ in both states. The individual's expected utility is $E[U] = \ln(0.1) - (0.01)^2 \approx -2.303$, which is far below zero, the enormous tax required to sustain full insurance under moral hazard makes everyone drastically worse off.

Now suppose that, in light of the moral hazard problem identified above, the government moves away from full insurance. It instead offers a *partial*-insurance program with $b = 3$. The government expects that this lower benefit will induce the individual to choose $s = \frac{1}{3}$. Call this **Program #3**.

(g) [2 points] What tax p makes Program #3 budget balanced, given the government's expectation that $s = \frac{1}{3}$?

$P = s \cdot b = \frac{1}{3} \times \$3 = \$1$

(h) [2 points] Verify the government's expectation: solve for the individual's optimal s under Program #3. Does the individual indeed choose $s = \frac{1}{3}$?

$$E[U] = (1-s)\ln(9) + s\ln(2) - (1-s)^2$$

while $w - p = 10 - 1 = 9$ and $b - p = 3 - 1 = 2$

$$\Rightarrow \frac{dE[U]}{ds} = -\ln(9) + \ln(2) + 2(1-s) = 0$$

$$\Rightarrow 2(1-s) = \ln(9) - \ln(2) = \ln\left(\frac{9}{2}\right) \approx 1.5041$$

$$\Rightarrow 1 - s \approx 0.7521$$

$$\Rightarrow s^* = 1 - \frac{\ln\left(\frac{9}{2}\right)}{2} \approx 0.248$$

Since $s^* \neq 0.333 \rightarrow$ not choose $s = \frac{1}{3}$

(i) [2 points] What is the individual's utility under Program #3?

Since $s^* \approx 0.248$. thus $1 - s^* \approx 0.752$

$$\Rightarrow E[U] = 0.752 \ln(9) + 0.248 \ln(2) - (0.752)^2$$

$$\Rightarrow \begin{cases} 0.752 \times \ln(9) = 0.752 \times 2.1972 \approx 1.652 \\ 0.248 \times \ln(2) = 0.248 \times 0.6931 \approx 0.172 \\ (0.752)^2 = 0.565 \end{cases}$$

$$\Rightarrow E[U] \approx 1.652 + 0.172 - 0.565 \approx 1.259$$

(j) [2 points] Given the individual's actual choice of s from part (h), does Program #3 run a surplus, break even, or run a deficit? Explain briefly.

Tax revenue $= p = \$1$

$\rightarrow$ Expected benefit payments $= s^* \cdot b = 0.248 \times \$3 = \$0.744$

$\rightarrow$ Budget $= 1 - 0.744 = +\$0.256$

$\rightarrow$ surplus of 0.256

P#3 runs a surplus of \$0.256. The government set $P = \$1$ expecting $s = \frac{1}{3}$, but the individual actually chose $s^* \approx 0.248 < \frac{1}{3}$. Because unemployment was lower than anticipated, benefit payments fell short of tax revenue, producing a surplus. The lower benefit did more to encourage job search than the government expected.

(k) [2 points] Compare Programs #2 and #3 from a social welfare perspective. Which program should the government prefer, and why? Your answer should explicitly discuss the trade-off between *consumption smoothing* (the benefit of insurance) and *moral hazard* (the cost of insurance), and draw a general lesson for the design of unemployment insurance. **Note:** If you did not successfully compute utility under Program #2, you may use the fact that it is less than 0.

The government should prefer P#3. Although P#2 provides full consumption smoothing, the moral hazard it generates is catastrophic: the individual chooses $s^* = 0.99$, requiring a tax of 9.9 to balance the budget, which leaves consumption at only 0.1 in each state, giving utility of -2.303. P#3 provides only partial insurance ($b = \$3$), but this preserves job-search incentives ($s^* \approx 0.248$), keeps the tax low ($p = \$1$), and delivers utility of approximately 1.259. The general lesson for UI design is that the optimal benefit is positive but below full insurance: full insurance eliminates the consumption smoothing benefit by requiring such high taxes that it defeats its own purpose, while partial insurance balances the gains from consumption smoothing against the costs of moral hazard.